Geometric Sequences and Series

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Geometric sequences**

For the specified parameter configuration, consider the sequence $3, 6, 12, 24, \ldots$

a State the common ratio. $r = 2$ **b** Write a formula for the nth term. $a_n = 3 \cdot 2^{n-1}$ **c** Find a_{10}. $a_{10} = 1536$

(a) $r = 2$; (b) $a_n = 3 \cdot 2^{n-1}$; (c) $a_{10} = 1536$

2 **[Foundational]** • **Target: Geometric sequences**

In a geometric sequence, $a_2 = 6$ and $a_5 = 48$. Find the common ratio and the first term.

$r^3 = \frac{48}{6} = 8$, so $r = 2$ and $a_1 = 3$.

3 **[Foundational]** • **Target: Geometric series**

Find the sum of the first 8 terms of $5 + 10 + 20 + \cdots$

$S_8 = 5\,\frac{2^8 - 1}{2 - 1} = 5(255) = 1275$.

4 **[Foundational]** • **Target: Geometric series**

Evaluate each infinite sum, or state that it diverges:

a $18 + 6 + 2 + \frac{2}{3} + \cdots$ $r = \frac{1}{3}$: $S = \frac{18}{1 - 1/3} = 27$.

b $1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} + \cdots$ $r = -\frac{1}{2}$: $S = \frac{1}{1 + 1/2} = \frac{2}{3}$.

c $2 + 3 + 4.5 + \cdots$ $r = 1.5 > 1$: diverges.

(a) $r = \frac{1}{3}$: $S = \frac{18}{1 - 1/3} = 27$.; (b) $r = -\frac{1}{2}$: $S = \frac{1}{1 + 1/2} = \frac{2}{3}$.; (c) $r = 1.5 > 1$: diverges.

5 **[Foundational]** • **Target: Geometric series**

Write the repeating decimal $0.\overline{7}$ as a fraction using an infinite geometric series.

$0.7 + 0.07 + \cdots$ with $a = \frac{7}{10}$, $r = \frac{1}{10}$: $S = \frac{7/10}{9/10} = \frac{7}{9}$.

6 **[Foundational]** • **Target: Geometric series**

A ball is dropped from 8 m and rebounds to $\frac{3}{4}$ of its previous height on each bounce. Find the total vertical distance it travels (falling and rising) before coming to rest.

Total $= 8 + 2\left(8 \cdot \frac{3}{4} + 8 \cdot \left(\frac{3}{4}\right)^2 + \cdots\right) = 8 + 2\,\frac{6}{1 - 3/4} = 8 + 48 = 56$ m.

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7 [Foundational] • Target: Geometric means and mixed sequences

Insert two positive geometric means between 4 and 108 (i.e., find b_1, b_2 so that $4, b_1, b_2, 108$ is geometric).

4 terms total: $r^3 = \frac{108}{4} = 27$, so $r = 3$. Means: $12, 36$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Geometric means and mixed sequences

Determine whether each sequence is arithmetic, geometric, or neither:

a $2, 6, 18, 54, \ldots$ Geometric ($r = 3$)

b $5, 8, 11, 14, \ldots$ Arithmetic ($d = 3$)

c $1, 4, 9, 16, \ldots$ Neither (differences $3, 5, 7$ are not constant, and ratios are not constant)

(a) Geometric ($r = 3$); (b) Arithmetic ($d = 3$); (c) Neither (differences $3, 5, 7$ are not constant, and ratios are not constant)

9 [Intermediate] • Target: Applications

A car worth 80,000 riyals depreciates by 15% per year.

a Write a formula for its value after n years. $V_n = 80000(0.85)^n$.

b Find its value after 5 years, to the nearest riyal. $V_5 = 80000(0.85)^5 \approx 80000(0.4437) \approx 35{,}497$ riyals.

(a) $V_n = 80000(0.85)^n$.; (b) $V_5 = 80000(0.85)^5 \approx 80000(0.4437) \approx 35{,}497$ riyals.

10 [Intermediate] • Target: Applications

A chain letter starts with 1 person sending it to 5 people, each of whom sends it to 5 more, and so on. Find the total number of people who have received the letter after 6 rounds (including the original sender's round).

This is a geometric series with $a = 1$, $r = 5$, $n = 6$: $S_6 = \frac{1(5^6 - 1)}{5 - 1} = \frac{15624}{4} = 3906$.

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11 **[Intermediate]** • **Target: Visualizing geometric growth**

The first 6 terms of $a_n = 2(1.5)^{n-1}$ are plotted below. Contrast the shape of this graph with the straight-line shape of an arithmetic sequence's graph.

Unlike an arithmetic sequence's straight-line pattern, the geometric sequence's points curve upward with increasing steepness — each term is a constant MULTIPLE of the previous one rather than a constant ADDITION, producing exponential-style curvature rather than a line.

12 **[Intermediate]** • **Target: Determining convergence before summing**

For each infinite geometric series, determine whether it converges WITHOUT computing the sum, by checking $|r| < 1$: $\sum_{n=1}^{\infty} 5(0.9)^{n-1}$, $\sum_{n=1}^{\infty} 3(1.2)^{n-1}$, $\sum_{n=1}^{\infty} 7(-0.4)^{n-1}$.

First: $r = 0.9$, $|r| < 1$: converges. Second: $r = 1.2$, $|r| > 1$: diverges. Third: $r = -0.4$, $|r| < 1$: converges.

13 **[Intermediate]** • **Target: Finding the number of terms from a partial sum**

For the geometric series $2 + 6 + 18 + \cdots$, find the number of terms n needed for the sum to first exceed 2000.

$S_n = \frac{2(3^n - 1)}{3 - 1} = 3^n - 1 > 2000 \Rightarrow 3^n > 2001$. Testing: $3^6 = 729$, $3^7 = 2187$. So $n = 7$ terms are needed (giving $S_7 = 3^7 - 1 = 2186 > 2000$).

14 **[Intermediate]** • **Target: A mixed application: population with geometric decline**

A population of an endangered species declines by 12% each year from an initial 5000. Find the total number of 'animal-years' lived by the population over the first 10 years (i.e. the sum of the population sizes at the start of each of the 10 years), to the nearest whole number.

$a_1 = 5000$, $r = 0.88$. $S_{10} = \frac{5000(1 - 0.88^{10})}{1 - 0.88} = \frac{5000(1 - 0.2785)}{0.12} = \frac{5000(0.7215)}{0.12} \approx 30{,}063$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Finding the number of terms from a partial sum** • **Cross-Lesson: 4.03 Hyperbolas**

An infinite geometric series has first term $a_1 = 18$ and sum to infinity $S_\infty = 24$. (a) Find the common ratio r. (b) Calculate the third term a_3.

(a) Find common ratio r: $S_\infty = \frac{a_1}{1 - r} \Rightarrow 24 = \frac{18}{1 - r} \Rightarrow 1 - r = \frac{18}{24} = \frac{3}{4} \Rightarrow r = 1 - \frac{3}{4} = \mathbf{\frac{1}{4}}$. (b) Calculate 3rd term a_3: $a_3 = a_1 \cdot r^2 = 18 \cdot \left(\frac{1}{4}\right)^2 = 18 \cdot \frac{1}{16} = \mathbf{\frac{9}{8} = 1.125}$.

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16 [Challenging] • Target: A mixed application: population with geometric decline • Cross-Lesson: 5.01 Vectors: Components and Operations

An infinite geometric series has first term $a_1 = 36$ and converges to $S_\infty = 48$. What is the common ratio r, and what is the third term a_3?

Using the infinite geometric sum formula $S_\infty = \frac{a_1}{1-r}$: $48 = \frac{36}{1-r} \Rightarrow 1 - r = \frac{36}{48} = \frac{3}{4} \Rightarrow r = 1 - \frac{3}{4} = \mathbf{\frac{1}{4}}$ Since $|r| = \frac{1}{4} < 1$, the convergence condition holds. Find the third term $a_3 = a_1 r^2$:
$a_3 = 36\left(\frac{1}{4}\right)^2 = 36 \times \frac{1}{16} = \mathbf{\frac{9}{4}} = 2.25$

17 [Challenging] • Target: Geometric sequences • Cross-Lesson: 5.02 The Dot Product

Evaluate the finite sum: $\sum_{k=1}^{6} 3(2)^{k-1}$

Step 1: Identify geometric series parameters: For $k = 1$, $a_1 = 3(2)^0 = 3$. The common ratio is $r = 2$ and the number of terms is $n = 6$. Step 2: Apply finite geometric sum formula:

$S_6 = \frac{a_1(1-r^6)}{1-r} = \frac{3(1-2^6)}{1-2} = \frac{3(1-64)}{-1} = \frac{3(-63)}{-1} = 3 \times 63 = \mathbf{189}$.

18 [Challenging] • Target: Geometric series • Cross-Lesson: 5.03 Parametric Equations

An infinite geometric series has first term $a_1 = 5$ and common ratio $r = 0.6$. Find its sum S_∞.

Since $|r| = |0.6| < 1$, the infinite geometric series converges. Apply $S_\infty = \frac{a_1}{1-r}$:
$S_\infty = \frac{5}{1-0.6} = \frac{5}{0.4} = \frac{5}{2/5} = \mathbf{\frac{25}{2}} = \mathbf{12.5}$

19 [Challenging] • Target: Geometric means and mixed sequences • Cross-Lesson: 6.01 Polar Coordinates and Conversions

Find the first three non-zero terms of the Maclaurin series for e^{2x}. State the range of validity.

$e^{2x} = 1 + 2x + \frac{(2x)^2}{2!} + \cdots = 1 + 2x + 2x^2 + \cdots$. Valid for all $x \in \mathbb{R}$.

20 [Challenging] • Target: Applications • Cross-Lesson: 6.02 Polar Equations and Graphs

Find the binomial expansion of $\frac{1}{\sqrt{4-x}}$ in ascending powers of x up to and including the term in x^2. State the set of values of x for which the expansion is valid.

Rewrite with negative fractional index and factor out 4: $(4-x)^{-1/2} = 4^{-1/2}\left(1 - \frac{x}{4}\right)^{-1/2} = \frac{1}{2}\left(1 - \frac{x}{4}\right)^{-1/2}$
Apply $(1+y)^n = 1 + ny + \frac{n(n-1)}{2}y^2 + \cdots$ with $n = -\frac{1}{2}$ and $y = -\frac{x}{4}$:
$\left(1 - \frac{x}{4}\right)^{-1/2} = 1 + \left(-\frac{1}{2}\right)\left(-\frac{x}{4}\right) + \frac{\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)}{2}\left(-\frac{x}{4}\right)^2 + \cdots$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Geometric sequences

How many terms of the geometric series $1 + 3 + 9 + \cdots$ give a sum of 364?

a. 5

b. 7

c. 6 ✓

d. 8

$\frac{3^n - 1}{2} = 364 \Rightarrow 3^n = 729 = 3^6 \Rightarrow n = 6.$

22 [Foundational] • Target: Geometric series

Find the number of terms in an AP with first term 5, last term 41, sum 230.

a. 10 ✓

b. 15

c. 12

d. 8

$S_n = n(5 + 41)/2 = 23n = 230 \Rightarrow n = 10.$

23 [Foundational] • Target: Geometric means and mixed sequences

The sum of the first n terms of a geometric series is $S_n = \frac{a_1(1 - r^n)}{1 - r}$. Find S_5 for $a_1 = 2, r = 3$.

a. 486

b. 243

c. 242 ✓

d. 240

$2(1 - 3^5)/(1 - 3) = 2(1 - 243)/(-2) = 242.$

24 [Foundational] • Target: Applications

The first term of a geometric progression is 16 and the fourth term is 2. Find the sum to infinity, S_∞, of the progression.

a. 32 ✓

b. $\frac{64}{3}$

c. 24

d. 30

The nth term of a geometric progression is given by $u_n = ar^{n-1}$, where a is the first term and r is the common ratio. Given $a = 16$ and the fourth term $u_4 = 2$: $ar^3 = 2 \Rightarrow 16r^3 = 2 \Rightarrow r^3 = \frac{2}{16} = \frac{1}{8} \Rightarrow r = \frac{1}{2}$ Since $|r| < 1$, the sum to infinity exists and is given by: $S_\infty = \frac{a}{1 - r} = \frac{16}{1 - \frac{1}{2}} = \frac{16}{\frac{1}{2}} = 32$ Common errors include setting $u_4 = ar^4$ which yields $r^4 = \frac{1}{8}$, or miscalcula.

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25 [Foundational] • Target: Visualizing geometric growth

If $|r| < 1$, prove $\sum_{n=0}^{\infty} r^n = \frac{1}{1-r}$ using the limit of partial sums.

a. Use calculus only

b. Cannot be proved

c. $S_N = \frac{1-r^{N+1}}{1-r} \to \frac{1}{1-r}$ **as** $r^{N+1} \to 0$ ✓

d. Only for $0 < r < 1$

Applying Sequences & Series principles, Classic geometric series proof. Therefore, the verified correct option is $S_N = \frac{1-r^{N+1}}{1-r} \to \frac{1}{1-r}$ as $r^{N+1} \to 0$.

26 [Foundational] • Target: Determining convergence before summing

Using partial fractions, find $\sum_{k=1}^{n} \frac{1}{(2k-1)(2k+1)}$.

a. 1

b. $\frac{1}{2}$

c. $\frac{n}{2n-1}$

d. $\frac{n}{2n+1}$ ✓

PFD: $\frac{1}{2}\left(\frac{1}{2k-1} - \frac{1}{2k+1}\right)$. Telescopes to $\frac{1}{2}\left(1 - \frac{1}{2n+1}\right) = \frac{n}{2n+1}$.

27 [Foundational] • Target: Finding the number of terms from a partial sum

A geometric series with $|r| < 1$ has an infinite sum of:

a. $S = a_1 \cdot r$

b. $S = \frac{a_1}{1-r}$ ✓

c. Undefined

d. $S = \frac{a_1}{r}$

Applying Sequences & Series principles, Infinite geometric series sum. Therefore, the verified correct option is $S = \frac{a_1}{1-r}$.

28 [Foundational] • Target: A mixed application: population with geometric decline

Use generating functions to find the number of ways to write n as an ordered sum of positive integers.

a. n^2

b. 2^{n-1} ✓

c. $n!$

d. 2^n

Applying Sequences & Series principles, Compositions of n: 2^{n-1}. Therefore, the verified correct option is 2^{n-1}.

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Geometric sequences**

Find the sum of an infinite geometric series where $a_1 = 6$ and $r = 2/3$.

a. 6

b. 12

c. 18 ✓

d. 9

Applying Sequences & Series principles, $S = 6/(1 - 2/3) = 6 \cdot 3 = 18$. Therefore, the verified correct option is 18.

30 **[Intermediate]** • **Target: Geometric series**

A geometric series has $a_1 = 100$ and $r = 0.9$. Find the infinite sum.

a. 900

b. 1000 ✓

c. 10

d. 100

Applying Sequences & Series principles, $S = 100/(1 - 0.9) = 1000$. Therefore, the verified correct option is 1000.

31 **[Intermediate]** • **Target: Geometric means and mixed sequences**

What is the sum of the first n terms of $1 + 1 + 1 + \cdots$?

a. n^2

b. n ✓

c. $n/2$

d. ∞

Applying Sequences & Series principles, n ones sum to n. Therefore, the verified correct option is n.

32 **[Intermediate]** • **Target: Applications**

For the arithmetic-geometric series $\sum_{k=1}^{\infty} kr^k$ with $|r| < 1$, find the sum.

a. $\frac{r}{(1-r)^2}$ ✓

b. $\frac{1}{(1-r)^2}$

c. $\frac{r}{1-r}$

d. $\frac{1}{1-r}$

Differentiate geometric series formula w.r.t. r: $\sum kr^{k-1} = 1/(1-r)^2 \Rightarrow \sum kr^k = r/(1-r)^2$.

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33 [Intermediate] • Target: Visualizing geometric growth

The geometric series $8 + 4 + 2 + 1 + \cdots$ has infinite sum:

a. 16 ✓

b. 32

c. 15

d. 8

Applying Sequences & Series principles, $S = 8/(1 - 1/2) = 16$. Therefore, the verified correct option is 16.

34 [Intermediate] • Target: Determining convergence before summing

The nth partial sum of a series is $S_n = n^2 - 3n$. Find the first term a_1.

a. 1

b. 2

c. -2 ✓

d. -3

Applying Sequences & Series principles, $a_1 = S_1 = 1 - 3 = -2$. Therefore, the verified correct option is -2.

35 [Intermediate] • Target: Finding the number of terms from a partial sum

Find the sum of the geometric series $1 + \frac{2}{3} + \frac{4}{9} + \cdots$ (infinite).

a. $\frac{3}{2}$

b. 3 ✓

c. 2

d. ∞

Applying Sequences & Series principles, $S = 1/(1 - 2/3) = 3$. Therefore, the verified correct option is 3.

36 [Intermediate] • Target: A mixed application: population with geometric decline

Find how many terms of the arithmetic series $5 + 8 + 11 + \cdots$ are needed for the sum to exceed 100.

a. $n = 7$

b. $n = 8$ ✓

c. $n = 10$

d. $n = 9$

$S_n = n(5 + (3n + 2)/2) = n(3n + 7)/2 > 100 \Rightarrow 3n^2 + 7n - 200 > 0 \Rightarrow n > 8.1 \Rightarrow n = 9$.

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37 [Intermediate] • Target: Geometric sequences

Determine convergence of $\sum_{n=1}^{\infty} \frac{n!}{n^n}$.

a. Converges to e

b. **Converges (ratio test: $a_{n+1}/a_n \to 1/e < 1$)** ✓

c. Inconclusive

d. Diverges

$\frac{(n+1)!/(n+1)^{n+1}}{n!/n^n} = \left(\frac{n}{n+1}\right)^n \to 1/e < 1.$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Geometric series • Cross-Lesson: 5.03 Parametric Equations

A geometric series has sum $S_n = 5(1 - 0.2^n)$. Identify a_1 and r.

a. $a_1 = 1,\ r = 0.2$

b. $a_1 = 4,\ r = 0.8$

c. $a_1 = 4,\ r = 0.2$ ✓

d. $a_1 = 5,\ r = 0.2$

$S_n = \frac{a_1(1-r^n)}{1-r}$. Here $1 - r = 1 - 0.2 = 0.8 \Rightarrow a_1 = 5 \cdot 0.8 = 4$.

39 [Challenging] • Target: Geometric means and mixed sequences • Cross-Lesson: 6.01 Polar Coordinates and Conversions

A geometric series has first term $a = 2$ and common ratio $r = -\frac{1}{4}$. Find the sum to infinity.

a. $\frac{8}{5}$ ✓

b. $\frac{18}{5}$

c. $\frac{5}{2}$

d. $-\frac{8}{5}$

$S_\infty = \frac{a}{1-r} = \frac{2}{1-(-\frac{1}{4})} = \frac{8}{5}$.

40 [Challenging] • Target: Applications • Cross-Lesson: 6.02 Polar Equations and Graphs

For the series $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$, find the partial fraction and sum.

a. ∞

b. $\frac{1}{2}$

c. 1 ✓

d. $\ln 2$

Telescoping: $\frac{1}{n} - \frac{1}{n+1}$. Sum telescopes to 1.

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41 [Challenging] • Target: Visualizing geometric growth • Cross-Lesson: 7.01 Exponential Functions and Their Graphs

The sum of the first n terms of a series is $S_n = n^2 + 3n$. Is the series arithmetic?

a. Yes, $d = 3$

b. No

c. **Yes,** $d = 2$ ✓

d. Cannot determine

$a_n = S_n - S_{n-1} = 2n + 2$. Linear in n: arithmetic with $d = 2$.

42 [Challenging] • Target: Determining convergence before summing • Cross-Lesson: 7.02 Logarithms and Their Properties

A geometric series has first term $a = 5$ and common ratio $r = 2$. Find S_3, the sum of the first 3 terms.

a. **35** ✓

b. 40

c. 75

d. 15

$S_3 = \frac{a(1-r^n)}{1-r} = \frac{5(1-(2)^3)}{1-(2)} = 35$

43 [Challenging] • Target: Finding the number of terms from a partial sum • Cross-Lesson: 7.03 Exponential and Logarithmic Equations

What is $\sum_{k=0}^{n} r^k$ (geometric sum, $req1$)?

a. $(n+1)r^n$

b. $\frac{r^n - 1}{r - 1}$

c. r^n

d. $\frac{r^{n+1} - 1}{r - 1}$ ✓

Sum k from 0 to n: one extra term compared to 1 to n.

44 [Challenging] • Target: A mixed application: population with geometric decline • Cross-Lesson: 7.04 Applications: Growth and Decay

Find the radius of convergence of $\sum_{n=0}^{\infty} \frac{x^n}{n!}$.

a. $R = e$

b. $R = \infty$ ✓

c. $R = 0$

d. $R = 1$

Ratio test: $|x|/(n+1) \to 0$. Converges for all x.

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45 [Challenging] • Target: Geometric sequences • Cross-Lesson: 8.01 Rational Functions and Asymptotes

A geometric series with common ratio $r = \frac{1}{3}$ has sum to infinity $S_\infty = 12$. Find the first term a.

a. 11

b. 7

c. **8** ✓

d. -8

$a = S_\infty(1 - r) = 12 \times \left(1 - (\frac{1}{3})\right) = 8.$

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